

Evaluating LLM-Based and Rule-Based Adaptive Mental Health Questionnaires: A Human-Centric User Study

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Abstract—

I. INTRODUCTION

Mental health is increasingly recognised as a critical component of overall well-being, and digital health applications are playing a growing role in routine screening and self-monitoring. In practice, most of these applications rely on standardised static instruments such as the PHQ-9 [1] and the GAD-7 [2], which present the same fixed set of items at every check-in regardless of the user's current condition, prior responses, or longitudinal history. While these instruments are clinically validated, their rigid format can feel repetitive and burdensome, and often fails to reflect how a person's mental state evolves over time [3].

Adaptive questionnaire techniques, such as branching logic and computerised adaptive testing, have been introduced to partially address this limitation by tailoring item sequences to user input [4]. However, these rule-based mechanisms remain bounded by predefined decision trees and cannot generate new items or reframe questions based on a user's individual context. Recent advances in Large Language Models (LLMs) open a new design space for adaptive assessment: LLMs can synthesise context-aware questions informed by a user's health profile and prior responses, and have begun to be explored as engines for patient-reported outcome measures [5].

Despite this promise, the use of LLMs in mental health remains at an early stage. A recent systematic review of mental health chatbots highlights that most existing systems are still rule-based, and that LLM-based designs, while more flexible, introduce new concerns around safety, appropriateness, and clinical reliability [6], [7], [8]. To our knowledge, no prior work has directly compared rule-based and LLM-based adaptive mental health questionnaires within a single unified application using a controlled, human-centric evaluation.

To address this gap, we developed *Navigator*, a web-based prototype that implements two distinct monthly mental health check-in methods within the same application environment:

- **Rule-based method:** predefined branching logic determines which follow-up questions appear based on fixed risk-level thresholds.
- **LLM-based method:** Google Gemini 2.5 Flash dynamically generates personalised questions from the user's health profile and prior assessment data.

Using a within-subjects study design, participants interacted with both methods via a standardised persona (Sophie) and completed a structured feedback survey. Evaluation metrics are grounded in the COSMIN content-validity framework [9], which identifies *Clarity*, *Quality*, and *Relevance* as core dimensions of questionnaire content validity.

This study aims to answer the following research questions:

- **RQ1 (Clarity):** Do LLM-generated questionnaires produce higher perceived clarity than rule-based questionnaires?
- **RQ2 (Quality):** Do LLM-generated questionnaires produce higher perceived quality than rule-based questionnaires?
- **RQ3 (Relevance):** Do LLM-generated questionnaires produce higher perceived relevance than rule-based questionnaires?

The main contributions of this work are: (1) the Navigator prototype, a Flutter web application implementing both rule-based and LLM-based adaptive questionnaires within a unified interface; (2) a within-subjects user study design using a standardised persona to ensure fair comparison across conditions; and (3) an empirical, COSMIN-grounded evaluation of user perceptions of Clarity, Quality, and Relevance for LLM-based versus rule-based adaptive mental health questionnaires.

The remainder of this paper is organised as follows. Section II reviews related work on mental health questionnaire systems, LLMs in healthcare, and adaptive survey design. Section III presents the methodology, covering the research approach, system design, controlled scenario, study design, and evaluation metrics. Section IV presents the results, followed by discussion in Section V, and conclusions in Section VI.

II. RELATED WORK

Although the broader Navigator project was initially framed around adaptive assessment across multiple chronic-disease domains [10], this paper narrows the evaluation to mental health check-ins, which provide a concrete, high-impact, and survey-feasible context for comparing rule-based and LLM-based adaptation methods. We accordingly organise prior work along three lines that progressively narrow toward our research question. We first review the dominant paradigm of static,

validated mental health instruments and the limitations they impose on longitudinal self-report. We then survey existing adaptive questionnaire strategies, from computerised adaptive testing to rule-based interface adaptation, and most recently to LLM-generated patient-reported outcome measures. Finally, we examine the emerging body of work on LLM-based mental health systems, identifying both their promise and the safety considerations that constrain their design. We conclude by introducing the COSMIN content-validity framework [9], which provides the three evaluation dimensions used in this study.

A. Mental Health Assessment and Static Instruments

Mental health screening in primary care and digital health applications has long relied on standardised, fixed-item self-report instruments. The Patient Health Questionnaire-9 (PHQ-9) for depression [1] and the Generalised Anxiety Disorder-7 (GAD-7) scale [2] remain among the most widely used tools, demonstrating strong sensitivity and specificity (both 88% for PHQ-9 ≥ 10 ; 89% and 82% for GAD-7 ≥ 10) and providing interpretable severity cut-offs across diverse populations. Their brevity, transparency, and validated scoring make them attractive for integration into digital platforms aimed at routine, longitudinal mental health monitoring.

However, the very stability that gives these instruments their psychometric Strength also imposes meaningful limitations on user experience. Static questionnaires repeat the same items at every administration, regardless of a user’s current state or response history. Repeated exposure to identical wording can lead to response fatigue and reduced engagement, particularly in longitudinal self-report scenarios where users complete the same scale weekly or monthly [11], [12], [3]. Ecological momentary assessment studies have documented this problem directly: in a 28-day smartphone-based depression study, only seven of eleven participants completed the full protocol and the average daily response rate was 62.5% [13], suggesting that Repeated digital self-report can suffer from engagement and completion issues challenges, even when the underlying measures are clinically meaningful.

Beyond engagement, fixed wording can raise concerns about content validity. Baas et al. [14] demonstrated that one PHQ-9 item exhibited measurement non-invariance across ethnic groups, revealing cultural bias in supposedly universal items. Wild et al. [15] similarly argues that questionnaire wording must be treated as a structured design and validation problem rather than a literal translation. These findings indicate that even well-validated instruments carry implicit assumptions about who the respondent is and how they interpret each item. Adaptive systems may help identify and mitigate these assumptions through more context-sensitive wording and item selection.

B. Adaptive Questionnaire Approaches

To mitigate the burden and rigidity of static instruments, prior work has explored adaptive questioning strategies. The most established approach is Computerised Adaptive Testing

(CAT), which uses item response theory to select The most informative items based on prior responses. Gibbons et al. [11] demonstrated that the CAT-based assessment of mood and anxiety disorders reduced the number of administered items by approximately 95% relative to a full item bank, while preserving strong correlation with the full-scale score. Choi et al. [12] likewise compared static short forms, two-stage branching, and CAT against the full PROMIS depression item bank, finding that CAT outperformed static short forms on most performance criteria, with two-stage semi-adaptive testing approaching CAT-level performance.

A parallel line of work has examined rule-based adaptive interfaces in broader health contexts. In a systematic review of adaptive user interfaces for chronic disease, Wang et al. [4] identify three dominant adaptation strategies: rule-based “if-then” logic, machine-learning-driven prediction, and feedback-loop systems that revise content based on accumulating user data. Although rule-based approaches dominate the surveyed literature due to Their interpretability and low computational overhead, they are bounded by The comprehensiveness of their predefined rules prevents them from adapting to contexts The designer did not anticipate. Closed-loop systems, which integrate continuous data collection with iterative refinement remains comparatively rare [16].

A more recent direction extends adaptation beyond item selection toward item generation. Terheyden et al. [5] introduces the concept of LLM-powered Patient-Reported Outcome Measures (LLM-PROMs), in which large language models generate personalised open-ended items and interpret responses against established psychometric scaffolds. Their framework argues that LLM-based PROMs should be developed from validated content and benchmarked against fixed instruments to preserve content validity. This conceptual shift, from *selecting* items in a finite bank to *generating* items conditioned on user context, defines the design space we examine in this paper. Unlike CAT, which adapts by selecting from a validated item bank, LLM-based methods can also alter wording or generate new follow-up prompts, creating both opportunities for personalisation and additional risks for validity.

C. LLM-Based Systems in Mental Health

The emergence of large language models has accelerated interest in conversational and adaptive mental health tools, but evidence on their efficacy and safety remains preliminary. In the most comprehensive review to date, Hua et al. [6] surveyed 160 mental health chatbot studies published between 2020 and 2024, classifying them as rule-based, machine-learning-based, or LLM-based. Their analysis reveals an important asymmetry: although LLM-based Systems grew rapidly in 2024, they remain concentrated in early-stage validation, while rule-based systems still dominate later-stage human and clinical testing. This positioning motivates our study as a pilot, user-centred feasibility evaluation rather than a clinical efficacy trial.

User-facing evidence further suggests that LLM behaviour in mental health is highly condition-dependent. Li et al. [7] analysed 112,698 social-media posts about LLM use for mental health and found that sentiment varied substantially across conditions: users with ADHD and ASD reported strongly positive experiences (sentiment scores of 0.722 and 0.557 respectively), whereas schizophrenia-spectrum discussions reported strongly negative sentiment (-0.569), often describing chatbot responses that appeared to intensify delusional thinking. Complementing this, Shen et al. [8] conducted a clinician-blind evaluation of multiple LLM versions on 158 prompts and found that no tested model reliably produced appropriate responses to psychotic content. Together, these findings establish two requirements for any LLM-based mental health tool: condition-aware framing and explicit safety boundaries. This distinction is also important for our setting: questionnaire systems do not aim to provide therapeutic dialogue, but rather to elicit structured, interpretable self-reports through clear and relevant prompts. The risks identified in conversational LLM use therefore translate, but do not transfer directly, into the design constraints for LLM-generated questionnaire items.

Despite this growing body of work, limited prior research has directly compared rule-based and LLM-based adaptive questionnaires within a unified application using a controlled, within-subjects evaluation. Existing comparisons either contrast different chatbot generations against one another [6], study LLM use in unstructured user discussions [7], or evaluate adaptive item *selection* without addressing LLM-generated *wording* [11], [12]. Our work addresses this gap by holding the application environment, the evaluation persona, and the participant group constant, thereby isolating the effect of the adaptation method itself.

D. Evaluation Framework: Content Validity via COSMIN

To evaluate the two adaptation methods on common ground, we adopt the COSMIN content-validity framework [9]. Drawing on an international Delphi study of 159 experts, COSMIN formalises content validity along three dimensions: *relevance*, *comprehensiveness*, and *comprehensibility*, for the target population and context of use. We map these directly onto the three evaluation metrics used in this paper: *comprehensibility* corresponds to *Clarity* (whether the wording is easy to understand), *relevance* corresponds to *Relevance* (whether the items fit the participant’s context), and *comprehensiveness* corresponds to perceived *Quality*, since participants are asked to judge whether each adaptive question appears complete, appropriate, and useful within the monthly mental-health check-in context (see Section III-H). This grounding provides a principled basis for comparing rule-based and LLM-based adaptive questionnaires that draws on established patient-reported-outcome methodology rather than ad-hoc usability heuristics.

III. METHODOLOGY

This section presents the end-to-end methodology used in this study. The research proceeded through six stages: (1) a literature review of static and adaptive mental health instruments to identify the limitations that motivate this work (Section II); (2) identification of validated questionnaire items, drawing primarily on the PHQ-9 [1] and GAD-7 [2], and a decision to focus the evaluation on the mental health domain; (3) design and implementation of two adaptation methods, rule-based and LLM-based, within a unified web-based prototype (Sections III-A–III-C); (4) construction of a controlled evaluation scenario using a standardised persona (Section III-D); (5) design of a within-subjects user study with a COSMIN-grounded feedback survey (Sections III-E–III-H); and (6) data collection and analysis (Section IV).

The prototype was designed as an evaluation instrument rather than a production system: its purpose is to present both methods within a single, controlled application environment so that user perceptions can be compared fairly. Architectural choices, including a shared interface, fixed persona, and identical health domains, were therefore driven by the evaluation design.

A. Architecture Overview

Navigator is implemented as a Flutter web application backed by Supabase (PostgreSQL) for question storage and branching-rule definitions. A single-page interface presents foundational intake questions, monthly check-ins for six health domains (Mental Health, Dietary, Physical Activity, Alcohol, Smoking/Vaping, and Women’s Health), and a results dashboard. Although the broader prototype supports all six domains, the present study evaluates only the Mental Health monthly check-in pathway, which provides a concrete and high-impact context for comparing adaptation methods. Both the rule-based and LLM-based monthly check-in methods share the same front-end layout, question rendering components, and scoring pipeline, so that any difference participants observe can be attributed to the adaptation method itself rather than to interface variation.

For LLM-based question generation, the application calls the Google Gemini 2.5 Flash API. This model was selected for its low latency and structured-output support (native JSON response mode), which are practical requirements for generating questionnaire items in real time within a web application. A deterministic fallback path returns pre-authored enhanced questions if the API is unavailable, ensuring that every participant in the study receives the same enhanced questionnaire content regardless of network conditions.

B. Rule-Based Method

The rule-based method adapts monthly check-in questionnaires through a branching engine that evaluates predefined conditional rules after each user response. Each rule is a triple of *trigger question*, *boolean expression*, and *target action*. When a user answers the trigger question, the engine evaluates the expression against the response value; if

the condition is met, the target questions are shown or hidden accordingly. This approach mirrors the “if-then” rule-based adaptation strategy identified as the dominant paradigm in chronic-disease adaptive interfaces [4].

Rules operate on single-question, single-timepoint data: a rule can reference the current answer to one question but cannot access prior month responses or aggregate longitudinal patterns. For the Mental Health domain, risk levels are classified at the domain level using project-specific percentage-based thresholds loosely inspired by the PHQ-9 severity bands [1]: scores below 25% of the domain maximum are classified as *None*, 25–50% as *Mild*, 50–75% as *Moderate*, and above 75% as *High*. These thresholds are a simplified mapping for the prototype and do not correspond directly to the validated PHQ-9 cut-off scores.

A concrete example illustrates the method’s limitation. Table I lists the mental health questions presented under each method in Sophie’s scenario (Section III-D). Under the rule-based method, Rule 13 specifies that the functional-impact question is shown only when the mood frequency response is ≥ 2 on a 0–3 scale. Sophie’s Month 2 response of 1 (“Several days”) does not meet this threshold, so the functional-impact item remains hidden, even though two consecutive months of Mild risk may warrant exploring it. The rule-based system cannot detect this longitudinal pattern because each rule evaluates a single response in isolation.

C. LLM-Based Method

The LLM-based method generates personalised monthly check-in questions by conditioning a large language model on the user’s prior health snapshot. At the start of a monthly check-in, the system constructs a structured prompt containing two inputs: (1) a *UserSeed*, a versioned summary of the user’s demographics, domain-level risk classifications, and free-text context notes from the previous assessment period; and (2) the baseline rule-based questions for the current domain, so the model understands what it is improving upon.

The prompt instructs the model to produce questions that are *more relevant, clearer, and less repetitive* than the rule-based baseline, grounding the generation task in the same content-validity dimensions (Clarity, Quality, Relevance) used in the evaluation (Section III-H). The model returns a JSON array of question objects, each containing the question text, response options, and a brief rationale for the adaptation. Generation uses a low temperature setting ($T = 0.3$) to favour consistency and structured output over creative variation.

This design aligns with the LLM-PROM framework proposed by Terheyden et al. [5], which argues that LLM-generated patient-reported outcome items should be developed from validated content and benchmarked against fixed instruments. In Navigator, the LLM does not generate questions from scratch; it *transforms* the existing validated baseline questions using the user’s longitudinal context. Table I presents the five specific transformations applied in Sophie’s scenario: history-aware rephrasing of the mood frequency item, occupation-specific language for the

manageability item, a new trend-direction item that captures the two-month persistent Mild pattern, early unlocking of the functional-impact item based on longitudinal rather than single-month thresholds, and omission of the isolation item where Month 1 data showed no change was expected.

D. Controlled Scenario: Sophie

To enable a fair within-subjects comparison, both methods are evaluated using a single standardised persona rather than participants’ own health data. “Sophie” is a fictional 24-year-old female East Asian international university student whose Month 1 foundational assessment produced a Mild mental health risk classification and None across all other domains. Her context notes record specific response details: a mood frequency score of 1 (“Several days”), a low and stable isolation score, and no triggered functional-impact questions.

The decision to use a fixed persona serves three evaluation purposes. First, it eliminates between-participant variation in health profiles, ensuring that every participant sees the same rule-based and LLM-based questionnaire outputs and that observed differences in Clarity, Quality, and Relevance ratings reflect the adaptation method rather than individual health circumstances. Second, it avoids the ethical complications of collecting real mental health data in a prototype evaluation context. Third, it provides a scenario that concretely illustrates the limitation of rule-based adaptation: Sophie’s profile is designed so that the branching rule’s hard threshold blocks the functional-impact question despite a clinically meaningful two-month pattern (see Table I), creating a clear point of contrast for participants to evaluate.

The persona was designed with attention to measurement validity considerations identified in prior work. Baas et al. [14] demonstrated that PHQ-9 responses can exhibit measurement non-invariance across demographic groups, and Wild et al. [15] emphasised that questionnaire wording must account for cultural context. Sophie’s profile, an East Asian international student, was therefore chosen to represent a demographic for whom culturally adapted wording (as produced by the LLM method) might be particularly relevant, while remaining relatable to the predominantly university-aged participant pool.

E. Research Design

This study uses a within-subjects design: every participant interacts with both the rule-based and the LLM-based monthly check-in methods and then rates each on the same set of evaluation items. The within-subjects approach was chosen because it allows each participant to serve as their own control, increasing statistical sensitivity to differences between methods while requiring fewer participants than a between-subjects design.

Participants completed the rule-based check-in first, followed by the LLM-based check-in. This fixed presentation order was adopted because the rule-based condition establishes a baseline context that participants can reference when evaluating the LLM-based adaptations; the LLM method is designed to improve upon the specific limitations of the

TABLE I
MENTAL HEALTH CHECK-IN QUESTIONS UNDER EACH ADAPTATION METHOD FOR SOPHIE’S MONTH 2 SCENARIO. THE RULE-BASED COLUMN SHOWS THE QUESTIONS PRODUCED BY THE BRANCHING ENGINE; THE LLM-BASED COLUMN SHOWS THE GEMINI-GENERATED TRANSFORMATIONS. DASHES INDICATE ITEMS NOT PRESENTED.

Item	Rule-Based	LLM-Based	Transformation Rationale
Mood frequency	“In the past month, how often have you felt low mood, sad, hopeless, nervous, worried, or restless?”	“Over the past month, have you continued to experience feelings like low mood, worry, or restlessness? Last month you reported this happening on several days.”	References Month 1 history; acknowledges continuity rather than repeating identical wording.
Managing life	“...how often have you felt that important things in your life were difficult to manage?”	“...has it felt harder than usual to stay on top of your studies, daily responsibilities, or relationships?”	Replaces generic “important things” with student-specific language.
Trend direction	Not asked	“Since you’ve reported similar feelings for two months in a row, how would you describe the change over the past month?”	New item capturing longitudinal signal (two-month Mild pattern) that rule-based thresholds cannot detect.
Functional impact	<i>Hidden</i> because the mood score of 1 does not meet the ≥ 2 threshold required by the branching rule.	“Have these feelings made it harder for you to manage your daily life, for example, attending classes, completing work, or maintaining relationships?”	Unlocked early based on persistent pattern; phrasing adapted to student context.
Isolation	“In the past month, how often do you feel isolated from others?”	<i>Skipped</i> because the Month 1 score was low and stable, so no new information was expected.	De-prioritises items unlikely to yield new signal, reducing respondent burden.

rule-based output (see Table I), and viewing the baseline first makes these improvements interpretable. Counterbalancing was not implemented. We acknowledge this as a limitation: order effects cannot be fully ruled out, and this is discussed further in Section V-B.

The study was approved by the Monash University Human Research Ethics Committee (MUHREC) under a Non-HREC (Lower Risk) review (Project ID 51664). All participants were fully informed of the study’s purpose and provided consent via a written explanatory statement before beginning.

F. Participants

Participants were recruited through convenience sampling via direct contact with peers, classmates, and personal networks. Eligible participants were adults aged 18 years or older, comprising university students and working professionals. No monetary compensation or reimbursement was offered. In accordance with the ethics approval, students enrolled at Monash University Malaysia and Indonesia campuses were excluded from recruitment, as the approval was scoped to the Australian campus.

G. Procedure

Each participant completed the study independently in a single session, following a four-step procedure:

- 1) **Orientation.** Participants received a link to the Navigator prototype along with a user guide explaining the application interface and the Sophie persona. The

guide described Sophie’s demographics, her Month 1 health snapshot, and the purpose of the monthly check-in. Participants were instructed to evaluate the questionnaire from Sophie’s perspective.

- 2) **Rule-based check-in.** Participants completed Sophie’s Month 2 mental health check-in using the rule-based method. The branching engine determined which follow-up questions appeared based on their responses to the pre-filled baseline items. Three questions were presented; the functional-impact item remained hidden because the branching rule’s threshold was not met (Table I).
- 3) **LLM-based check-in.** Participants then completed the same Month 2 check-in using the LLM-based method, which presented the transformed questions generated from Sophie’s longitudinal profile. The interface layout and interaction pattern were identical to the rule-based condition.
- 4) **Feedback survey.** After completing both check-ins, participants were directed to an external Google Form survey comprising five sections: *Section A* (Demographics) collected age range, gender, health-app usage frequency, and student/professional status; *Section B* (Clarity), *Section C* (Relevance), and *Section D* (Quality) each contained three 5-point Likert items, where every item was rated for both the rule-based and LLM-based methods side by side; *Section E* (Overall Experience) asked participants to

indicate their overall preference among four options (rule-based, LLM-based, both, or neither) and to provide free-text feedback. Sections B and D also included an open-ended question each, yielding three open-ended prompts in total.

The entire session, including prototype interaction and survey completion, was estimated to take 10–15 minutes.

H. Evaluation Metrics

Evaluation metrics are grounded in the three content-validity dimensions identified by the COSMIN framework [9], mapped to the study context as follows:

- **Clarity** (COSMIN *comprehensibility*): whether the questionnaire wording is clear and easy to understand.
- **Quality** (COSMIN *comprehensiveness*): whether the questions are well-formed, meaningful, and complete within the monthly check-in context.
- **Relevance** (COSMIN *relevance*): whether the questions are appropriate and pertinent to Sophie's health profile and circumstances.

Each dimension was measured using three Likert-scale items rated from 1 (*Strongly Disagree*) to 5 (*Strongly Agree*). Each item was presented as a side-by-side matrix in which participants rated both the rule-based and LLM-based methods on the same screen, yielding nine paired ratings (18 data points) per participant. The three items within each dimension were averaged to produce a single composite score for Clarity, Quality, and Relevance for each method.

In addition, three open-ended questions were embedded across the survey sections: one within the Clarity section (asking about confusing or unclear questions), one within the Quality section (asking participants to compare how each version felt to complete), and one in the Overall Experience section (inviting general feedback on both methods). These qualitative responses supplement the Likert data by surfacing themes that structured items may not capture. A final preference item asked participants to select among four options: rule-based method, LLM-based method, both, or neither.

IV. RESULTS AND ANALYSIS

A. Participant Demographics

B. Quantitative Results

C. Qualitative Findings

D. Comparative Analysis

V. DISCUSSION

A. Interpretation of Results

B. Limitations

VI. CONCLUSION

Future Work

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